

# Display LaTeX tables in html and pdf at a one go

```
```{r setup}
library(qs); library(data.table); library(ggplot2)
library(rmarkdown); library(fixest)
library(knitr); library(tufte)
library(litedown)
litedown::reactor(
  eval = TRUE,
  echo = TRUE,
  time = TRUE
)
options(htmltools.dir.version = FALSE, width = 100)
#### threads
MaxCores <- parallel::detectCores()
setDTthreads(threads = MaxCores - 1)
setFixest_nthreads(nthreads = MaxCores - 1, save = TRUE)
#### ggplot theme
ThisTheme <- ggplot2::theme(
  axis.text.x = element_text(angle = 90, vjust = .5, hjust = .5),
  axis.text.y = element_text(size = 12),
  axis.title = element_text(size = 10),
  strip.text.x = element_text(color = "blue", size = 8,
    margin = margin(0, 1.25, 0, 1.25, "cm")),
  strip.text.y = element_text(color = "blue", size = 8,
    margin = margin(1.5, 0, 1.5, 0, "cm")),
  panel.spacing.x = unit(c(.1), units = "cm"),
  panel.spacing.y = unit(.1, units = "cm")
)
ThisThemeEnd <- ThisTheme + ggplot2::theme(legend.position="bottom")
```

```

#### invalidate cache when the tufte version changes
knitr::opts_chunk$set(
  tidy = FALSE, cache.extra = packageVersion('tufte'),
  margin_references = TRUE,
  #### remove leading hashes in html output
  comment = "",
  echo = T, cache = F, eval = T,
  class.source = "SeiroBenign", class.output = "SeiroLightGreen"
)
grepout <- function(str, x)
  # returns element of match (not numbers)
  x[grep(str, x, perl = T)]
table0 <- function(x) table(x, useNA = "ifany")
read.tablev <- function(fn, quotemark = "", headertrue = TRUE,
  blskip = TRUE, mat = F, ...)
{
  x <- read.table(fn, sep = "\t", header = headertrue, blank.lines.skip = blskip,
    quote = quotemark, na.strings = "", ...)
  if (is.null(dim(x)) | nrow(x) == 1)
  {
    cn <- names(x)
    dim(x) <- c(1, length(x))
    colnames(x) <- cn
  }
  if (mat) x <- as.matrix(x)
  return(x)
}
GeoLevel <- c("Prv", "Dis", "Loc", "Mai")
geovars <- c("busprov_geo", "busdistmuni_geo",
  "buslocmuni_geo", "busmainplc_geo")
...

```

```

~~~{r path}
path <- "W:/epguest/seiro_ito/"
pathprogram <- paste0(path, "outfiles/")
pathdata <- paste0(path, "data/")
pathsaveddata <- paste0(path, "saveddata/")
pathresults <- paste0(path, "results/")
pathdataCITIRP <- "U:/CIT-IRP5 Panel/"
pathdataIRP <- "U:/IRP5/Job level/v5/beta/"

```

...

```
```{r read functions}
destat <- function(x, prob = "short", signif = 1, removezero = F)
# returns an array of descriptive statistics
# prob specifies quantiles, "short", "even", "odd", "very short", "long"
{
  if (is.null(dim(x))) dim(x) <- c(length(x),1)
  if (is.null(colnames(x))) colnames(x) <- paste("v", 1:ncol(x), sepv = "")
  if (!is.data.frame(dim(x))) x <- data.frame(x)
  if (is.character(prob))
  {
    if (prob=="short")
    {
      prob <- c(.25,.5,.75); y <- array(dim=c(ncol(x),10))
      dimnames(y) <- list(colnames(x),c("min","25\\%", "median","75\\%", "max",
                                         "mean","std","0s","NAs","n"))
    } else if (prob=="even")
    {
      prob <- c(.2,.4,.5,.6,.8)
      y <- array(dim=c(ncol(x),12))
      dimnames(y) <- list(colnames(x),c("min","20\\%", "40\\%", "median","60\\%",
                                         "80\\%", "max","mean","std","0s","NAs","n"))
    } else if (prob=="odd")
    {
      prob <- c(.1,.25,.5,.75,.9)
      y <- array(dim=c(ncol(x),12))
      dimnames(y) <- list(colnames(x), c("min","10\\%", "25\\%", "median","75\\%",
                                         "90\\%", "max","mean","std","0s","NAs","n"))
    } else if (prob=="long")
    {
      prob <- c(.025,.1,.25,.5,.75,.9,.975)
      y <- array(dim=c(ncol(x),14))
      dimnames(y) <- list(colnames(x), c("min","2.5\\%",
                                         "10\\%", "25\\%", "median","75\\%", "90\\%", "97.5\\%",
                                         "max","mean","std","0s","NAs","n"))
    } else if (prob=="very short")
    {
      prob <- .5
      y <- array(dim=c(ncol(x),8))
    }
  }
}
```

```

    dimnames(y) <- list(colnames(x),c("min","median","max","mean","std","0s","NAs","n"))
  }
}
logical <- sapply(x, is.logical)
if (any(logical)) x[logical] <- lapply(x[logical], as.numeric)
char <- sapply(x, is.character)
if (any(char)) x[char] <- lapply(x[char], as.factor)
for (i in 1:ncol(x))
{
  z <- x[,colnames(x)[i]]
  zn <- na.omit(z)
  if (removezero) zn <- z[!(is.na(z) | (!is.na(z) & z == 0))]
  NAs <- sum(is.na(z))
  if (any(grepl("POSIX|Date", class(z))) | is.factor(z))
    y[i, ] <- c(rep(NA, 5 + length(prob)), NAs, length(z)) else
    y[i, ] <- c(round(c(min(zn, na.rm=T), quantile(zn, probs=prob), max(zn, na.rm=T),
                      mean(zn, na.rm=T), var(zn, na.rm=T)^(.5)), signif),
                sum(z==0, na.rm=T), NAs, length(z))
}
return(y)
}

destatfactor <- function(x, prob = "short", signif = 1, removezero = F, IgnoreIfAbove = 10)
# returns an array of descriptive statistics for factors/characters at all levels
# If gender = M, F is a factor, it gives 2 rows of gender: M, gender: F
# prob: specifies quantiles, "short", "even", "odd", "long","very short"
# IgnoreIfAbove: = 10, if length(levels(factor variable)) > 10, just skip
{
  if (is.null(dim(x))) dim(x) <- c(length(x),1)
  if (is.null(colnames(x))) colnames(x) <- paste("v", 1:ncol(x), sepv = "")
  if (!is.data.table(x)) x <- data.table(x)
  ##### if (!is.data.frame(dim(x))) x <- data.frame(x)
  iillogical <- sapply(x, is.logical)
  if (any(iillogical)) stop("Use destat for logical variables")
  iinum <- sapply(x, is.numeric)
  iiInt <- sapply(x, is.integer)
  if (any(iinum|iiInt)) {
    message(
      paste("Use destat for integer/numeric variables. Dropped:",
            paste(colnames(x)[iinum|iiInt], collapse = ", ")
    )
  }
}

```

```

    )
  )
  xd <- x[, !(iinum|iiInt), with = F]
} else xd <- x
iic <- sapply(xd, is.character)
iif <- sapply(xd, is.factor)
iifc <- iic|iif
#### Drop characters/factors with too many elements (> IgnoreIfAbove)
if (any(iifc)) {
  Numunq <- unlist(lapply(xd[, ..iifc], function(z) length(unique(z))))
  if (!is.null(Numunq)) ii100 <- Numunq > IgnoreIfAbove else ii100 <- F
  if (any(ii100)) {
    message(paste(names(xd[, ..iifc])[ii100], if (length(ii100)==1) "has" else "have",
      "more than", IgnoreIfAbove, "unique entries, dropped.\n"))
    #### drop iifc[ii100] columns, turn the rest to factors
    xf <- xd[, -which(iifc[ii100]), with = F]
    iic2 <- sapply(xf, is.character)
    xf[, (names(xf)[iic2]) :=
      lapply(.SD, as.factor), .SDcols = names(xf)[iic2]
    ]
  } else { # if number of all character/factor element <= IgnoreIfAbove
    xf <- xd
    xf[, (names(xf)[iifc]) :=
      lapply(.SD, as.factor), .SDcols = names(xf)[iifc]
    ]
  }
} else xf <- xd
#### list all levels
iif2 <- sapply(xf, is.factor)
AllLvList <- lapply(xf[, ..iif2], levels)
AllLvls <- unlist(AllLvList)
AllNames <- unlist(lapply(1:length(AllLvList),
  function(i) paste0(names(AllLvList)[i], ": ", AllLvList[[i]])))
Nrows <- length(AllLvls)
if (is.character(prob)) {
  if (prob=="short")
  {
    prob <- c(.25,.5,.75); y <- array(dim=c(Nrows, 10))
    dimnames(y) <- list(AllNames, c("min", "25\\%", "median", "75\\%", "max",
      "mean", "std", "0s", "NAs", "n"))
  }
}

```

```

} else if (prob=="even")
{
  prob <- c(.2,.4,.5,.6,.8)
  y <- array(dim=c(Nrows, 12))
  dimnames(y) <- list(AllNames, c("min","20\\%", "40\\%", "median", "60\\%",
                                   "80\\%", "max", "mean", "std", "0s", "NAs", "n"))
} else if (prob=="odd")
{
  prob <- c(.1,.25,.5,.75,.9)
  y <- array(dim=c(Nrows, 12))
  dimnames(y) <- list(AllNames, c("min","10\\%", "25\\%", "median", "75\\%",
                                   "90\\%", "max", "mean", "std", "0s", "NAs", "n"))
} else if (prob=="long")
{
  prob <- c(.025,.1,.25,.5,.75,.9,.975)
  y <- array(dim=c(ncol(x),14))
  dimnames(y) <- list(colnames(x), c("min","2.5\\%",
                                       "10\\%", "25\\%", "median", "75\\%", "90\\%", "97.5\\%",
                                       "max", "mean", "std", "0s", "NAs", "n"))
} else if (prob=="very short")
{
  prob <- .5
  y <- array(dim=c(Nrows, 8))
  dimnames(y) <- list(AllNames, c("min", "median", "max", "mean", "std", "0s", "NAs", "n"))
}
}

#### ..prefix allows data.table to search global environ, a level up from function
for (ii in 1:ncol(xf[, ..iif2])) {
  z <- xf[, colnames(xf[, ..iif2])[ii], with = F]
  zn <- na.omit(z)
  if (removezero) zn <- z[!(is.na(z) | (!is.na(z) & z == 0))]
  NAs <- sum(is.na(z))
  if (any(grepl("POSIX|Date", class(z))))
    y[ii, ] <- c(rep(NA, 5 + length(prob)), NAs, length(z)) else
  {
    Lvlis <- levels(unlist(zn))
    #### number of rows by far = length of AllLvList[1:(ii-1)]
    rowsbyfar <- if (ii > 1) length(unlist(AllLvList[1:(ii-1)])) else 0
    #### number of rows to add = length(AllLvList[ii])
    rows2add <- length(unlist(AllLvList[[ii]]))
  }
}

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    for (ll in 1:rows2add) {
      znI <- as.integer(zn == Lvls[ll])
      y[rowsbyfar+ll, ] <- c(round(c(
        min(znI, na.rm=T),
        quantile(znI, probs=prob, na.rm = T),
        max(znI, na.rm=T),
        mean(znI, na.rm=T),
        var(znI, na.rm=T)^(.5)),
        signif),
        sum(znI==0L, na.rm=T),
        NAs,
        sum(znI))
    }
  }
}
rownames(y) <- AllNames
return(y)
}

latextab.simple <- function(tab, hleft = NULL, hcenter = NULL, unit = "cm",
  hright = NULL, hline = NULL, cline = F,
  longtable = F, delimiterline = NULL,
  headercolor = NULL, alternatercolor = NULL,
  alternatercolor2 = NULL, alternaterstart2 = 3,
  alternatercolor4 = NULL, alternaterstart4 = 5,
  adjustlineskip = NULL)
# produces a LaTeX table.
#   hleft = header left: >{ "here" }
#   hcenter = header center length: p{ "here" lengthunit}, need for
#   unit = header center length unit: p{ length "here" }
#   hright = header right: <{ "here" }
#   hline = T: inserts "\hline" after each line
#           = numeric vector: inserts "\hline" at specified rows
#   cline = matrix(row, from, to) that gives "\cline{from-to}" at specified rows
#   longtable: if T, produce header/footer for longtable
#   delimiterline: if needed, set delimiterline = T which gives vertical lines
#   headercolor: if not NULL, specify color of the header row
#   alternatercolor: if not NULL, specify color of the alternating rows (1, 3, ...)
#   alternatercolor2: if not NULL, specify color of the alternating 2 rows (3:4, 6:7, ...)
#   alternatercolor4: if not NULL, specify color of the alternating 4 rows (5:8, 13:16, ...)

```

```

#   adjustlineskip: insert specified string to every second ""\\[here]""
{
  tab0 = copy(tab)
  if (!is.matrix(tab)) tab <- as.matrix(tab)
  ltab <- coln <- NULL
  if (is.null(hcenter)) hcenter <- rep(2, ncol(tab))
  if (length(hcenter) == 1) hcenter <- rep(hcenter, ncol(tab))
  for (i in 1:ncol(tab))
  {
    ltab <- cbind(ltab, tab[, i], " & ")
    coln <- c(coln,
      paste("\\makebox[", hcenter[i], unit, "{", colnames(tab)[i], "}", sep = "", collapse =
        , " & ")
    )
  }
  ltab <- rbind(coln, ltab)
  linebreak <- rep("\\\\", nrow(ltab))
  if (!is.null(adjustlineskip))
    linebreak[seq(2, nrow(ltab), 2)] <-
      paste(linebreak[seq(2, nrow(ltab), 2)], "[", adjustlineskip, "]", sep = "")
  if (!is.null(alternatecolor))
    linebreak[seq(2, nrow(ltab), 2)] <-
      paste(linebreak[seq(2, nrow(ltab), 2)], "\\rowcolor{", alternatecolor, "}", sep = "")
  if (!is.null(alternatecolor2))
  {
    # block sequence: 4, 5, 8, 9, ..., nrow(ltab)
    seqby2 <- rep(seq(alternatestart2, nrow(ltab), 4), each=2)
    seqby2[seq(2, length(seqby2), 2)] <- seqby2[seq(1, length(seqby2), 2)] + 1
    if (seqby2[length(seqby2)] > nrow(ltab)) seqby2 <- seqby2[-length(seqby2)]
    linebreak[seqby2] <-
      paste(linebreak[seqby2], "\\rowcolor{", alternatecolor2, "}", sep = "")
  }
  if (!is.null(alternatecolor4))
  {
    # block sequence: 5:8, 13:16, ..., nrow(ltab)
    seqby4 <- rep(seq(alternatestart4, nrow(ltab), 8), each = 4)
    for (k in 2:4)
      seqby4[seq(k, length(seqby4), 4)] <- seqby4[seq(1, length(seqby4), 4)] + k-1
    while (seqby4[length(seqby4)] > nrow(ltab)) seqby4 <- seqby4[-length(seqby4)]
    linebreak[seqby4] <-
      paste(linebreak[seqby4], "\\rowcolor{", alternatecolor4, "}", sep = "")
  }
}

```



```

}
if (is.null(hline)) holine <- "" else holine <- "\\hline"
linebreak <- paste(linebreak, holine, sep = "")
ltab[, ncol(ltab)] <- linebreak
# cline[, 1] + 1 because of the header line
if (!is.null(dim(cline)))
{
  if (nrow(ltab) == cline[nrow(cline), 1])
  {
    cat("\\cline cannot be added at the bottom row...ignored.")
    cline <- cline[-nrow(cline), , drop = FALSE]
  }
  clineadd <- rep("", nrow(ltab))
  clineadd[cline[, 1] + 1] <- paste("\\cline{", cline[, 2], "-", cline[, 3], "}", sep = "")
  # cline[, 1] + 1 because of the header line
  ltab[, ncol(ltab)] <- paste(ltab[, ncol(ltab)], clineadd, sep = "")
}
if (!is.null(hline)) ltab[c(1, nrow(ltab)), ncol(ltab)] <- "\\\\\\\\hline"
ltab2 <- NULL
for (i in 1:nrow(ltab)) ltab2 <- rbind(ltab2, paste(ltab[i, ], collapse = ""))
if (is.null(delimiterline)) delim <- "" else delim <- "|"
if (!(longtable))
{
  if (!is.null(hleft))
    head <- paste(
      c("\\begin{tabular}{",
        t(cbind(paste(delim, ">{", hleft, "}", sep = "")
          , paste("p{", hcenter, unit, "}", sep = "")
          , paste("<{", hright, "}", sep = "")))
        , paste0(delim, "}"))
      , collapse = "") else
    head <- paste(
      c("\\begin{tabular}{",
        t(cbind(paste(delim, ">{\\scriptsize\\hfil", rep("", ncol(tab)), "}", sep = "")
          , paste("p{", hcenter, unit, "}", sep = "")
          , paste("<{", rep("", ncol(tab)), "}", sep = "")))
        , paste0(delim, "}"))
      , collapse = "")
  foot <- "\\end{tabular}"
  if (!is.null(headercolor)) head <- paste0(head, "\\rowcolor{", headercolor, "}")

```

```

} else {
  if (!is.null(hleft))
    head1 <- paste(
      c("\\begin{longtable}{",
        t(cbind(paste(delim, ">{", hleft, "}", sep = "")
          , paste("p{", hcenter, unit, "}", sep = "")
          , paste("<{", rep("", ncol(tab)), "}", sep = "")))
        , paste0(delim, "}"))
      , collapse = "") else
    head1 <- paste(
      c("\\begin{longtable}{",
        t(cbind(paste(delim, ">{\\scriptsize\\hfil", rep("", ncol(tab)), "}", sep = "")
          , paste("p{", hcenter, unit, "}", sep = "")
          , paste("<{", rep("", ncol(tab)), "}", sep = "")))
        , paste0(delim, "}"))
      , collapse = "")
    head2 <- paste(paste(ltab[1, ], collapse = ""), "\\endfirsthead")
    head3 <- paste(paste(ltab[1, ], collapse = ""), "\\endhead")
    if (!is.null(headercolor)) {
      head2 <- paste0("\\rowcolor{", headercolor, "}", head2)
      head3 <- paste0("\\rowcolor{", headercolor, "}", head3)
    }
    ltab2 <- ltab2[-1, ]
    head <- rbind(head1, head2, head3)
    foot <- "\\end{longtable}"
  }
  if (!is.null(hline))
    ltab3 <- rbind(head, "\\hline", ltab2, foot) else
    ltab3 <- c(head, ltab2, foot)
  if (any(iidrop <- which(ltab3 == ""))) ltab3 <- ltab3[-iidrop, , drop = F]
  rownames(ltab3) <- NULL
  return(ltab3)
}

write.tablev <- function(x, fn,
  colnamestrue = TRUE, rownamestrue = FALSE,
  nastrings = "", ...)
  write.table(x, fn, sep = "\t", quote = FALSE,
    row.names = rownamestrue, col.names = colnamestrue, na = nastrings, ...)
...

```

## Method 1: define once, tex\_preview for html, cat(ltx) under results = asis for pdf

By saving the LaTeX to an R variable, you only have to write it once. Then you use two different code chunks to conditionally output for HTML vs PDF.

```
```{r define_table}
tab_string="
\\begin{tabular}{llr}
\\hline
\\multicolumn{2}{c}{Item} \\ \\ \\
\\cline{1-2}
Animal      & Description & Price (\\$) \\ \\ \\
\\hline
Gnat        & per gram    & 13.65      \\ \\ \\
            & each       & 0.01       \\ \\ \\
Gnu         & stuffed    & 92.50      \\ \\ \\
Emu         & stuffed    & 33.33      \\ \\ \\
Armadillo   & frozen     & 8.99       \\ \\ \\
\\hline
\\end{tabular}"
```
```

```
```{r render_html, results="asis", warning=FALSE, message=FALSE}
#| eval: !expr knitr::is_html_output()
library(texPreview)
tex_preview(tab_string, returnType = "html")
```
```

```
```{r render_pdf, results="asis"}
#| eval: !expr knitr::is_latex_output()
# For PDF, we just dump the raw LaTeX string into the document
cat(tab_string)
```
```

| Item      |             |            |
|-----------|-------------|------------|
| Animal    | Description | Price (\$) |
| Gnat      | per gram    | 13.65      |
|           | each        | 0.01       |
| Gnu       | stuffed     | 92.50      |
| Emu       | stuffed     | 33.33      |
| Armadillo | frozen      | 8.99       |

## Method 2: Produce image file with `tex_Preview` for html, use raw LaTeX contents for pdf

If you prefer to write the raw LaTeX block directly into the document instead of an R string, you must wrap it in ````\{=latex}` so that Quarto knows to completely hide it from the HTML version.

| Item      |             |            |
|-----------|-------------|------------|
| Animal    | Description | Price (\$) |
| Gnat      | per gram    | 13.65      |
|           | each        | 0.01       |
| Gnu       | stuffed     | 92.50      |
| Emu       | stuffed     | 33.33      |
| Armadillo | frozen      | 8.99       |

```
```\{r evaluate latex in html}
#| eval: !expr knitr::is_html_output()
#| out.width: "800px"
#| results: "hide"
tex_preview(latex_code,
  usrPackages = c('\\usepackage{booktabs}', '\\usepackage{amssymb}'),
  returnType = 'html',
  density = 300,
  fileDir = "c:/data/MinWageMarketPower/analysis/draft/",
  #####fileDir = getwd(),
  keep_pdf = T,
  imgFormat = "jpg",
  stem = "est201002w")
```\{`
```

```

```{r getwd}
getwd()
```

```

```
[1] "C:/data/MinWageMarketPower/analysis/draft"
```

FIGURE 1: RESULTS: 2010-2022, ESTABLISHMENTS WITH BASELINE JOBS GREATER THAN 2 BY HHI

### Job impacts: Low and high concentration, winsorized data

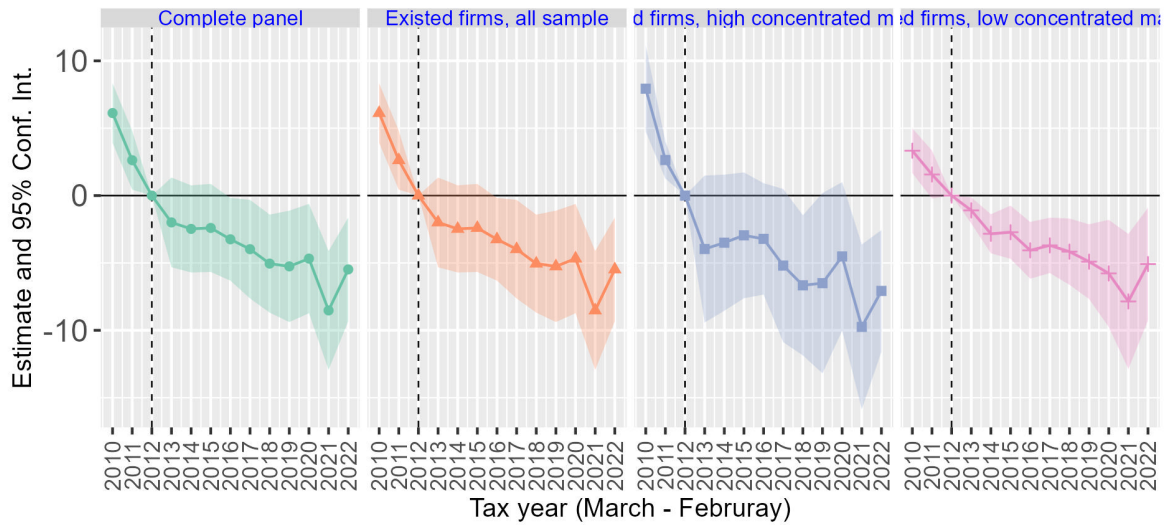


FIGURE 2: RESULTS: 2010-2022, ESTABLISHMENTS WITH BASELINE JOBS GREATER THAN 2 BY FA

### Job impacts: FA class in pre-2013, winsorized data

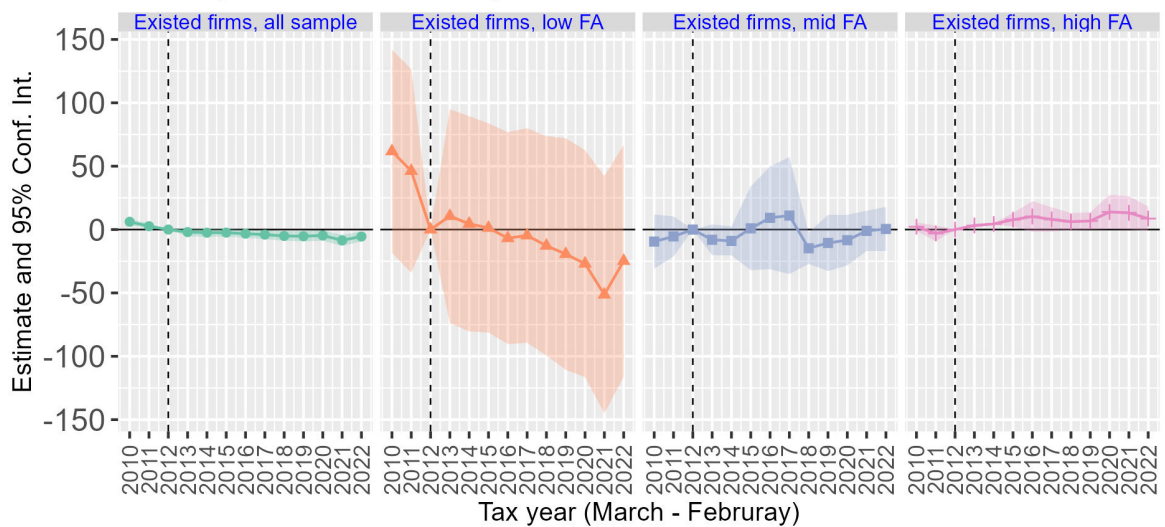


TABLE 1: RESULTS: 2010-2022, ESTABLISHMENTS WITH BASELINE JOBS GREATER THAN 2

|                                      | rJobsMP             |                      |                      |                      |                       |                   |                     |                      |
|--------------------------------------|---------------------|----------------------|----------------------|----------------------|-----------------------|-------------------|---------------------|----------------------|
|                                      | (1)                 | (2)                  | (3)                  | (4)                  | (5)                   | (6)               | (7)                 | (8)                  |
| Pre2013FAMP0 $\times$ taxyear = 2010 | 8.835***<br>(2.305) | 6.125***<br>(1.014)  | 6.125***<br>(1.014)  | 7.933***<br>(1.465)  | 3.331***<br>(0.7629)  | 61.77<br>(36.79)  | -9.496<br>(9.815)   | 2.233<br>(1.522)     |
| Pre2013FAMP0 $\times$ taxyear = 2011 | 4.114*<br>(2.178)   | 2.626**<br>(1.004)   | 2.626**<br>(1.004)   | 2.640***<br>(0.6336) | 1.565*<br>(0.8186)    | 46.25<br>(36.86)  | -5.413<br>(7.235)   | -3.065<br>(2.836)    |
| Pre2013FAMP0 $\times$ taxyear = 2013 | -1.644<br>(2.281)   | -1.991<br>(1.528)    | -1.991<br>(1.528)    | -3.966<br>(2.503)    | -1.096**<br>(0.4457)  | 10.62<br>(38.76)  | -8.058<br>(5.557)   | 3.173***<br>(0.6645) |
| Pre2013FAMP0 $\times$ taxyear = 2014 | -1.412<br>(2.706)   | -2.473<br>(1.484)    | -2.473<br>(1.484)    | -3.497<br>(2.318)    | -2.835***<br>(0.6657) | 4.518<br>(38.97)  | -8.976<br>(5.240)   | 4.396***<br>(0.2432) |
| Pre2013FAMP0 $\times$ taxyear = 2015 | -3.204<br>(2.829)   | -2.402<br>(1.499)    | -2.402<br>(1.499)    | -2.951<br>(2.143)    | -2.720**<br>(0.9053)  | 1.289<br>(37.87)  | 0.9528<br>(15.17)   | 7.759***<br>(2.463)  |
| Pre2013FAMP0 $\times$ taxyear = 2016 | -1.461<br>(2.958)   | -3.244**<br>(1.410)  | -3.244**<br>(1.410)  | -3.213<br>(1.898)    | -4.064***<br>(0.9671) | -6.806<br>(38.37) | 9.241<br>(18.61)    | 10.31*<br>(5.535)    |
| Pre2013FAMP0 $\times$ taxyear = 2017 | -3.638<br>(3.125)   | -3.978**<br>(1.679)  | -3.978**<br>(1.679)  | -5.195*<br>(2.609)   | -3.697***<br>(0.9420) | -4.660<br>(38.83) | 11.04<br>(21.16)    | 7.980*<br>(4.402)    |
| Pre2013FAMP0 $\times$ taxyear = 2018 | -3.565<br>(3.205)   | -5.049**<br>(1.669)  | -5.049**<br>(1.669)  | -6.661**<br>(2.388)  | -4.161***<br>(1.129)  | -12.78<br>(39.72) | -14.77**<br>(5.680) | 6.244*<br>(3.090)    |
| Pre2013FAMP0 $\times$ taxyear = 2019 | -3.219<br>(3.166)   | -5.254**<br>(1.897)  | -5.254**<br>(1.897)  | -6.501*<br>(3.070)   | -4.906***<br>(1.273)  | -19.34<br>(41.92) | -10.56<br>(10.17)   | 6.772*<br>(3.171)    |
| Pre2013FAMP0 $\times$ taxyear = 2020 | -1.202<br>(3.521)   | -4.676**<br>(1.861)  | -4.676**<br>(1.861)  | -4.516*<br>(2.529)   | -5.776***<br>(1.827)  | -26.91<br>(41.10) | -8.365<br>(9.151)   | 13.87**<br>(6.224)   |
| Pre2013FAMP0 $\times$ taxyear = 2021 | -3.925<br>(3.838)   | -8.523***<br>(2.023) | -8.523***<br>(2.023) | -9.746***<br>(2.804) | -7.855***<br>(2.298)  | -51.35<br>(42.94) | -1.057<br>(7.283)   | 13.22**<br>(6.021)   |
| Pre2013FAMP0 $\times$ taxyear = 2022 | -1.845<br>(3.716)   | -5.480***<br>(1.766) | -5.480***<br>(1.766) | -7.077***<br>(2.074) | -5.082**<br>(1.920)   | -24.78<br>(41.99) | 0.4629<br>(7.979)   | 8.661*<br>(4.384)    |
| R <sup>2</sup>                       | 0.61821             | 0.63593              | 0.63593              | 0.67566              | 0.61562               | 0.63219           | 0.59962             | 0.69126              |
| Observations                         | 40,873              | 26,292               | 26,292               | 13,009               | 12,511                | 7,827             | 3,729               | 14,695               |
| taxyear fixed effects                | ✓                   | ✓                    | ✓                    | ✓                    | ✓                     | ✓                 | ✓                   | ✓                    |
| Firm ID fixed effects                | ✓                   | ✓                    | ✓                    | ✓                    | ✓                     | ✓                 | ✓                   | ✓                    |
| Main Place fixed effects             | ✓                   | ✓                    | ✓                    | ✓                    | ✓                     | ✓                 | ✓                   | ✓                    |

```

```{r ltx}
ltx <- "\\begin{tabular}{lccccccc}
  \\toprule
  & \\multicolumn{8}{c}{rJobsMP}\\
  & (1) & (2) & (3) & (4) & (5) & (6) & (7) & (8)
  \\midrule
  Pre2013FAMP0 $\\times$ taxyear $=$ 2010 & 8.835$^{***}$ & 6.125$^{***}$ & 6.125$^{***}$ & 7.933$^{***}$ & 3.331$^{***}$ & 61.77 & -9.496 & 2.233
  & (2.305) & (1.014) & (1.014) & (1.465) & (0.7629) & (36.79) & (9.815) & (1.522)
  Pre2013FAMP0 $\\times$ taxyear $=$ 2011 & 4.114$^{*}$ & 2.626$^{**}$ & 2.626$^{**}$ & 2.640$^{***}$ & 1.565$^{*}$ & 46.25 & -5.413 & -3.065
  & (2.178) & (1.004) & (1.004) & (0.6336) & (0.8186) & (36.86) & (7.235) & (2.836)
  Pre2013FAMP0 $\\times$ taxyear $=$ 2013 & -1.644 & -1.991 & -1.991 & -3.966 & -1.096$^{**}$ & 10.62 & -8.058 & 3.173$^{***}$
  & (2.281) & (1.528) & (1.528) & (2.503) & (0.4457) & (38.76) & (5.557) & (0.6645)

```

```

& (2.281) & (1.528) & (1.528) &
Pre2013FAMPO $\times$ taxyear $=$ 2014 & -1.412 & -2.473 & -2.473 &
& (2.706) & (1.484) & (1.484) &
Pre2013FAMPO $\times$ taxyear $=$ 2015 & -3.204 & -2.402 & -2.402 &
& (2.829) & (1.499) & (1.499) &
Pre2013FAMPO $\times$ taxyear $=$ 2016 & -1.461 & -3.244$^{**}$ & -3.244$^{**}$ &
& (2.958) & (1.410) & (1.410) &
Pre2013FAMPO $\times$ taxyear $=$ 2017 & -3.638 & -3.978$^{**}$ & -3.978$^{**}$ &
& (3.125) & (1.679) & (1.679) &
Pre2013FAMPO $\times$ taxyear $=$ 2018 & -3.565 & -5.049$^{**}$ & -5.049$^{**}$ &
& (3.205) & (1.669) & (1.669) &
Pre2013FAMPO $\times$ taxyear $=$ 2019 & -3.219 & -5.254$^{**}$ & -5.254$^{**}$ &
& (3.166) & (1.897) & (1.897) &
Pre2013FAMPO $\times$ taxyear $=$ 2020 & -1.202 & -4.676$^{**}$ & -4.676$^{**}$ &
& (3.521) & (1.861) & (1.861) &
Pre2013FAMPO $\times$ taxyear $=$ 2021 & -3.925 & -8.523$^{***}$ & -8.523$^{***}$ &
& (3.838) & (2.023) & (2.023) &
Pre2013FAMPO $\times$ taxyear $=$ 2022 & -1.845 & -5.480$^{***}$ & -5.480$^{***}$ &
& (3.716) & (1.766) & (1.766) &

\\\\
R$^2$ & 0.61821 & 0.63593 & 0.63593 &
Observations & 40,873 & 26,292 & 26,292 &

\\\\
taxyear fixed effects & $\checkmark$ & $\checkmark$ & $\checkmark$ &
Firm ID fixed effects & $\checkmark$ & $\checkmark$ & $\checkmark$ &
Main Place fixed effects & $\checkmark$ & $\checkmark$ & $\checkmark$ &
\\bottomrule
\\end{tabular}
"
---
```

```

```{r headerrows}
headerrows <- c("& All sample & Complete panel & \\multicolumn{6}{c}{Incumbent firms} \\\\",
```
```

```

```{r evaluate latex in html2}
#| eval: !expr knitr::is_html_output()
#| out.width: "800px"
#| results: "hide"
library(texPreview)
```

```

tex_preview(ltx,
  usrPackages = c('\\usepackage{booktabs}', '\\usepackage{amssymb}'),
  returnType = 'html',
  density = 300,
  resizebox = T,
  fileDir = "c:/data/MinWageMarketPower/analysis/draft/",
  keep_pdf = T,
  imgFormat = "jpg",
  stem = "est201002w2")
```

```

```

```{r evaluate latex in html3, eval= F}
#| out.width: "800px"
#| results: "hide"
library(texPreview)
tex_preview(ltx,
  usrPackages = c('\\usepackage{booktabs}', '\\usepackage{amssymb}'),
  returnType = 'html',
  density = 300,
  resizebox = F,
  fileDir = "c:/data/MinWageMarketPower/analysis/draft/",
  keep_pdf = T,
  imgFormat = "jpg",
  stem = "est201002w_2")
```

```